

CAPSTONE PROJECT PROPOSAL

Preventing Data Breaches and Record Inaccuracies in Public Libraries Through a Cybersecurity-Enhanced Management System

In partial fulfillment for the requirements in Capstone 1 and Research

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Cyber Security Track

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CHAPTER 1

INTRODUCTION

1.1 Background Of The Study

Public libraries play an essential role in providing access to information, educational resources, and community services for students and the general public (Singh & Kumar, 2025). As technology continues to evolve, many libraries are transitioning from manual record keeping to digital management systems to improve accessibility and operational efficiency (Swaminathan, 2024). This shift allows libraries to enhance service delivery, streamline transactions, and provide faster access to information resources.

However, despite this shift toward digitization, numerous public libraries continue to face serious challenges related to cybersecurity and record accuracy. Many existing systems, especially those that are outdated or poorly maintained, lack proper security mechanisms, making them vulnerable to data breaches, unauthorized access, and data manipulation (Lewis & Lipkowitz, 2025). Sensitive information, such as borrower records, transaction history, and inventory details, can be exposed when security controls are weak or improperly implemented (Raghavaiah & Karnati, 2025).

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In addition, manual data entry and the absence of validation mechanisms frequently lead to inaccurate records, duplication of data, and inconsistencies within the system. These issues not only compromise the integrity of the library's information but also disrupt library operations, resulting in misplaced books, incorrect borrower information, and unreliable reporting. Therefore, there is a growing need for a secure and reliable library management system that integrates cybersecurity measures and data validation techniques to ensure both data protection and record accuracy.

Given these challenges, there is a clear need for a Cybersecurity-Enhanced Library Management System that integrates secure authentication, role-based access control, encryption, audit trails, and data validation mechanisms to protect sensitive information and improve record accuracy (Akor, 2024; Purdue University, 2025). By combining cybersecurity principles with structured system development, such a system can serve as a practical application of knowledge in software engineering, database management, and information security.

1.2 Statement of the Problem

Despite the availability of modern library technologies, many libraries still rely on manual or partially automated systems in managing records and daily transactions. These outdated practices result in inefficiencies, inaccuracies, and security vulnerabilities.

This study aims to address the following problems:

1. Library records and transactions are still managed manually, leading to slow processing and operational inefficiencies.
2. There is difficulty in accurately tracking borrowed and returned books, resulting in misplaced or unmonitored inventory.
3. Manual recording increases the risk of data inaccuracies, duplication, and inconsistencies in library records.
4. Existing systems lack adequate cybersecurity measures, making them vulnerable to unauthorized access, data breaches, and data manipulation.
5. There is no reliable backup and recovery mechanism, increasing the risk of data loss in case of system failure or cyberattacks.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop a secure and efficient web-based Library Management System that enhances inventory tracking, ensures accurate record management, and integrates cybersecurity measures to protect library data.

1.3.2 Specific Objectives

This study aims to:

1. To develop a secure and user-friendly web-based application interface that improves system accessibility and usability.
2. Implement cybersecurity measures to protect library records.
3. To design a data backup system to prevent loss of records.
4. Improve inventory tracking and monitoring of books in the library.
5. Provide real-time visibility of book availability for borrowing.

1.4 Scope and Delimitation

Scope:

This study focuses on the design and development of a web-based Cybersecurity-Enhanced Library Management System intended for public libraries in Iloilo and nearby communities. The system includes key functionalities such as user authentication, role-based access control (RBAC) for librarians and borrowers, book inventory management, and borrowing and return processes.

In addition, the system integrates cybersecurity features including data encryption, audit trails, real-time book availability checking, and automated data backup to ensure data protection and system reliability. The application is designed to be accessible to elementary, high school, and college users, providing a user-friendly interface for efficient library transactions.

Delimitation:

This study is limited to the design, development, and testing of a cybersecurity-enhanced system for managing digital library records within a single pilot public library. It does not include physical infrastructure or hardware-based security measures such as CCTV systems, biometric devices, or network firewalls.

The study excludes the management of rare books, special collections, and other non-digitized materials that require manual handling. Additionally, large-scale deployment,

long-term system performance evaluation, and nationwide implementation are beyond the scope of this study.

1.5 Significance of the Study

The proposed system will benefit the following:

- **Students** (elementary, high school, college)

The system allows students to easily check the availability of books in real time, reducing the time and effort required to search for resources and improving their overall learning experience. .

- **Local residents of Iloilo and nearby communities**

Community members will benefit from improved access to library information, enabling them to check available materials without physically visiting the library.

- **Librarians**

The system helps librarians manage records more efficiently by automating inventory tracking and borrowing processes. It reduces manual workload, minimizes errors, and enhances service delivery through a more organized and secure system.

- **Future Researchers**

This study may serve as a reference for future researchers interested in cybersecurity, web-based systems, and library management system development. It provides insights into integrating security features such as RBAC, encryption, and audit trails into practical applications.

1.6 Definition of Terms

Library Management System - A digital system used to manage library operations such as cataloging, borrowing, and returning of books.

Audit Trails - Records of all user activities within the system used to monitor actions and detect unauthorized access

Authentication - Process of verifying the identity of a user before granting access to the system

Borrower Record - Information of library users, consist of personal information and history of borrowing

Cybersecurity - A practice of protecting systems, networks, and data from cyber threats such as unauthorized access and data breaches

Database - A structured collection of data where book records, user information, and library transactions are stored.

Data Backup – A security feature that saves copies of library records to prevent data loss.

Data Breach - Incident where sensitive information is accessed without authorization

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Data Manipulation - Alteration or tampering of data, that affects accuracy and reliability

Data Validation - Ensuring that the data entered is accurate and in correct format.

Database Management - A process of implementing and maintaining databases to store and retrieve information efficiently.

Encryption - Changes data into a secure format to prevent unauthorized access.

Information Security - Protects information from unauthorized access, disclosure, disruption, modification, or destruction.

Real-Time Availability - System ability to instantly display the current status of books, either available or borrowed.

Role-Based Access Control (RBAC) - A security control that limits access based on roles

Software Engineering - The practice of designing, developing, testing and maintaining software applications properly.

Transaction History - Records of borrowing and returning activities that library uses do.

Unauthorized Access - An entry in a system or data without permission that leads to data breaches or manipulation.

Web-Based - Refers to a system that runs on a web browser, accessible via the internet.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Documents devastating ransomware attacks on major libraries like the British Library and Toronto Public Library, where entire systems including ILS, catalogs, and digital collections collapsed, preventing access to physical materials for months (Breeding, 2024). These incidents highlight vulnerabilities in library infrastructure that enable data encryption and exposure of sensitive patron information, emphasizing the critical need for robust cybersecurity defenses. The study urges libraries to implement regular software updates, isolated backups, dedicated security personnel, and staff training against social engineering to prevent such breaches and ensure operational continuity.

2.2 Local Studies

The study by Stephanie L. Lampaza ,Mary Joy P. Parrenas ; Angele D. Suarez ; Kristine T. Soberano,. (2025) showed that a Library Management System (LMS) improved library

operations by making processes faster, more accurate, and easier to use. Students, teachers, and librarians benefited from better access to information. However, some challenges were found, such as technical issues, lack of training, and resistance to using the new system. Other local studies also highlight that while libraries are shifting to digital systems, problems like weak security and inaccurate records still exist.

2.3 Related Literature

A Library Management System (LMS) is an information system that automates processes such as cataloging, borrowing, and returning of books, improving accuracy and reducing manual errors. Cybersecurity concepts such as authentication, encryption, and access control are used to protect systems and data from unauthorized access, ensuring data integrity, confidentiality, and security. The system is developed using modern web technologies including Python Flask, MySQL, and HTML, CSS, and JavaScript to create a secure and efficient web-based application for library management systems.

2.4 Synthesis of the Reviewed Literature

Foreign studies document devastating ransomware attacks crippling major libraries' ILS, catalogs, and digital collections for months (Breeding, 2024), while local studies reveal Philippine libraries struggle with technical failures, weak security, and inaccurate records despite

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LMS adoption (Lampaza et al., 2025). Both converge on vulnerabilities enabling data breaches and operational disruptions, recommending encryption, role-based access control, audit trails, regular backups, and staff training as essential defenses (Aregbesola & Nwaolise, 2023; Breeding, 2024). Related literature confirms libraries face escalating cyber threats but lack affordable, practical security implementations for resource-constrained environments.

Research Gap: Global frameworks exist but overlook Philippine public libraries' budget limitations and technical capacity. Local studies identify security weaknesses without providing accessible, student-developer-friendly solutions for small libraries like those in Iloilo. No reviewed studies offer simple web-based systems combining inventory management with borrower data protection tailored for under resourced institutions.

How Current Project Addresses Gaps:

This project delivers a web based inventory availability checker with borrower data security implementing proven safeguards encryption, role-based access, audit trails accessible to students and public libraries . Unlike complex enterprise solutions, it provides immediate, practical protection against ransomware and breaches (Breeding 2024), resolving local security gaps while respecting Philippine budget constraints.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study will use a developmental research design, which focuses on the systematic design, development, and evaluation of the proposed web application. This approach will ensure that the system is effectively developed and assessed based on its functionality, usability, and performance.

3.2 System Development Methodology

The System will be developed using Agile Development Methodology ,a flexible and iterative approach that allows continuous improvement throughout the development process. Agile focuses on collaboration, adaptability, and delivering functional components in incremental stages. This methodology is suitable for the project as it enables frequent feedback, quick adjustments, and efficient problem-solving.

The Agile Development Model consists of the following phases:



1. Plan - In this phase, the project requirements are identified and analyzed. The development team collaborates with stakeholders to define system goals, features, and scope. A project plan is created, including timelines, tasks, and resource allocation.

2. Design - The system architecture and interface are designed based on the identified requirements. This includes creating wireframes, database structures, and system flow diagrams to visualize how the system will function.

3. Develop - During this phase, the actual coding and implementation of the system take place. Developers build the features based on the design specifications while ensuring code quality and functionality.

4. Test - The system is tested to identify and fix errors or bugs. Various testing methods such as unit testing, integration testing, and user testing are conducted to ensure the system meets the required standards and performs correctly.

5. Deploy - After successful testing, the system is deployed and made available to users. This phase includes installation, configuration, and initial user support.

6. Review - In the final phase, feedback is gathered from users and stakeholders. The development team evaluates the system's performance and identifies areas for improvement. These insights are used for the next iteration, making Agile a continuous cycle.

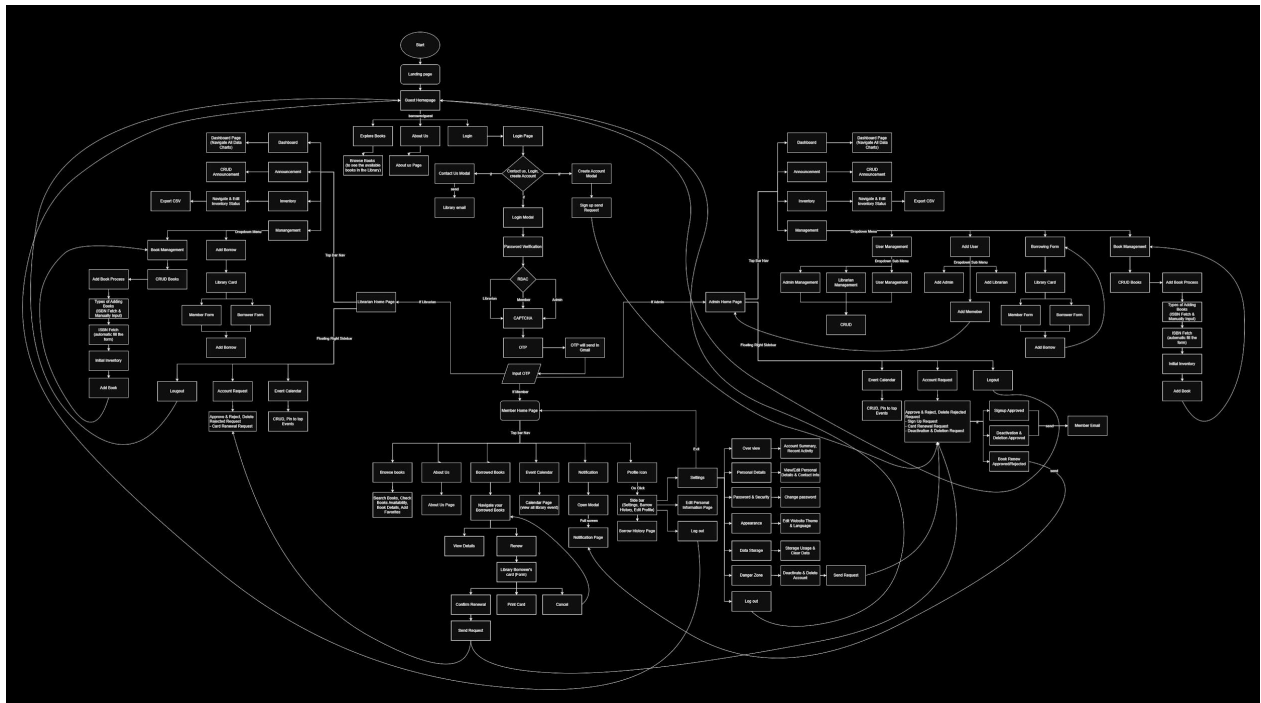
3.3 System Architecture

The system operates as a secure 3-role web application:

- Members/Users (view books availability).
- Librarians (manage records).
- Admins (full system control).

All interactions flow through Python Flask backend enforcing cybersecurity controls before accessing MySQL database. Data encrypted via HTTPS with role-based access control (RBAC).

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Client Server Set up

Client Browser (HTML/CSS/JS) Users	Web Server (Python + Apache) (Login Security)	Xampp MySQL Database (Data)
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Client: Any web browser accessing the system

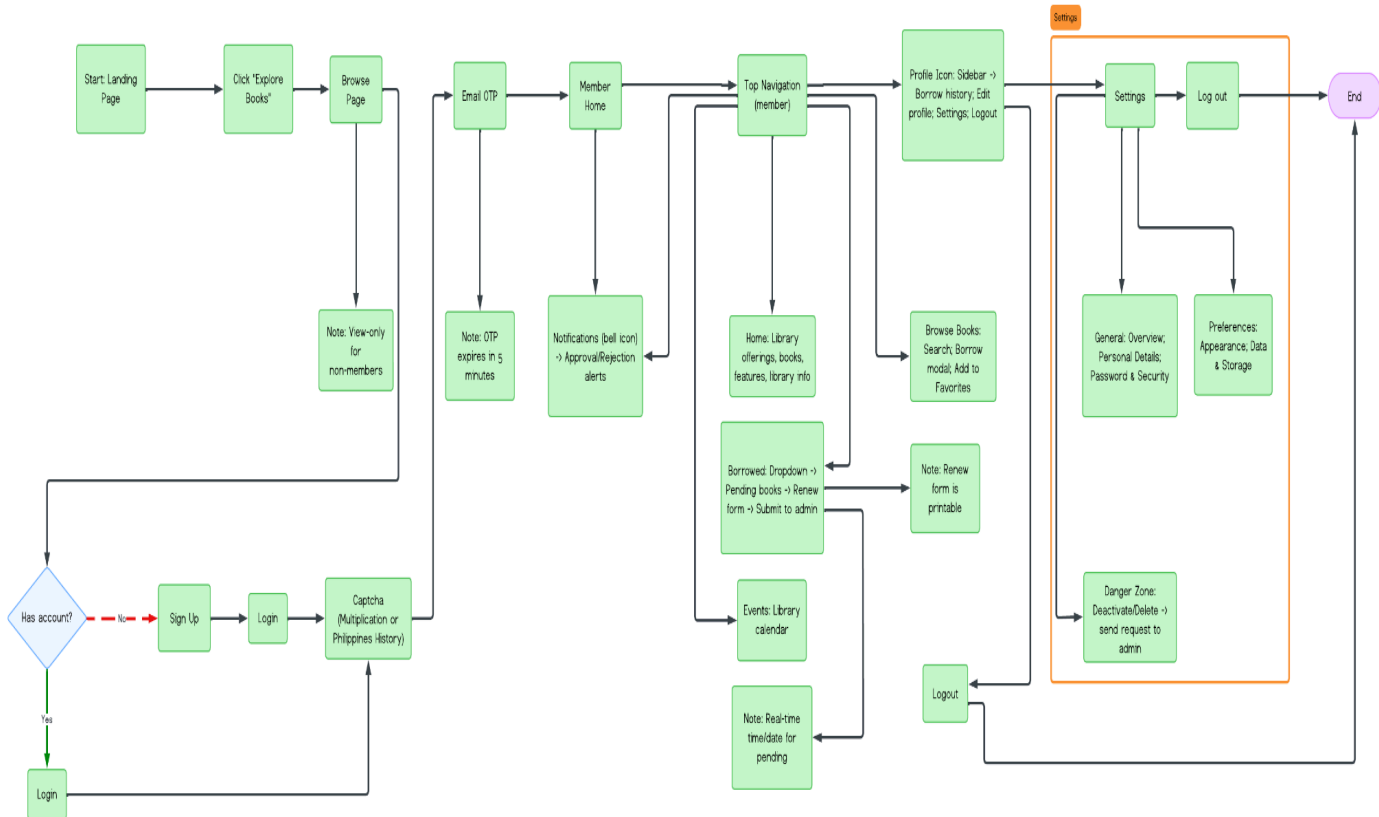
Server: localhost serves static files (HTML/CSS/JS)

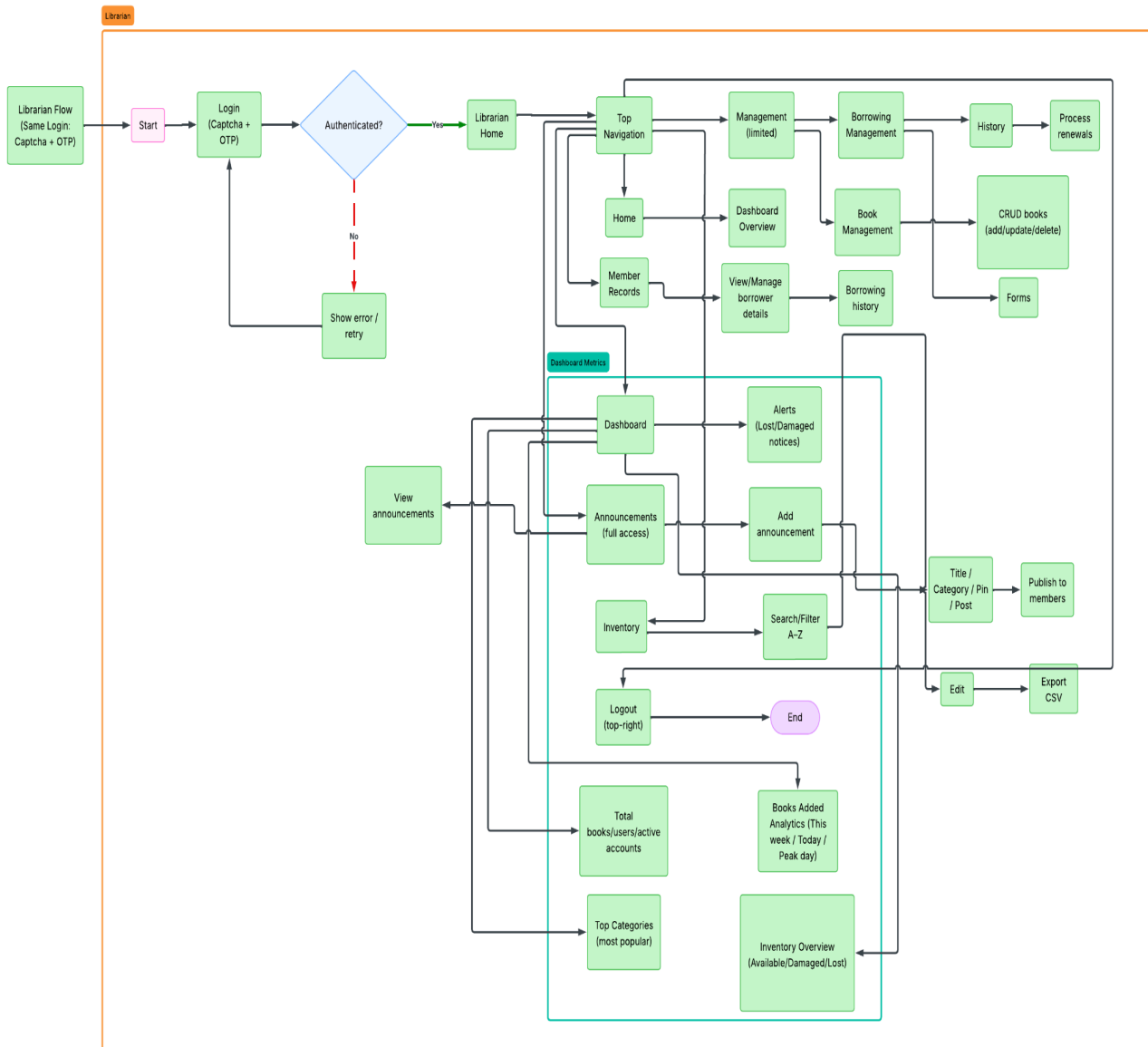
Database: Stores books, borrowers, transactions, audit logs

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User Interaction Flow

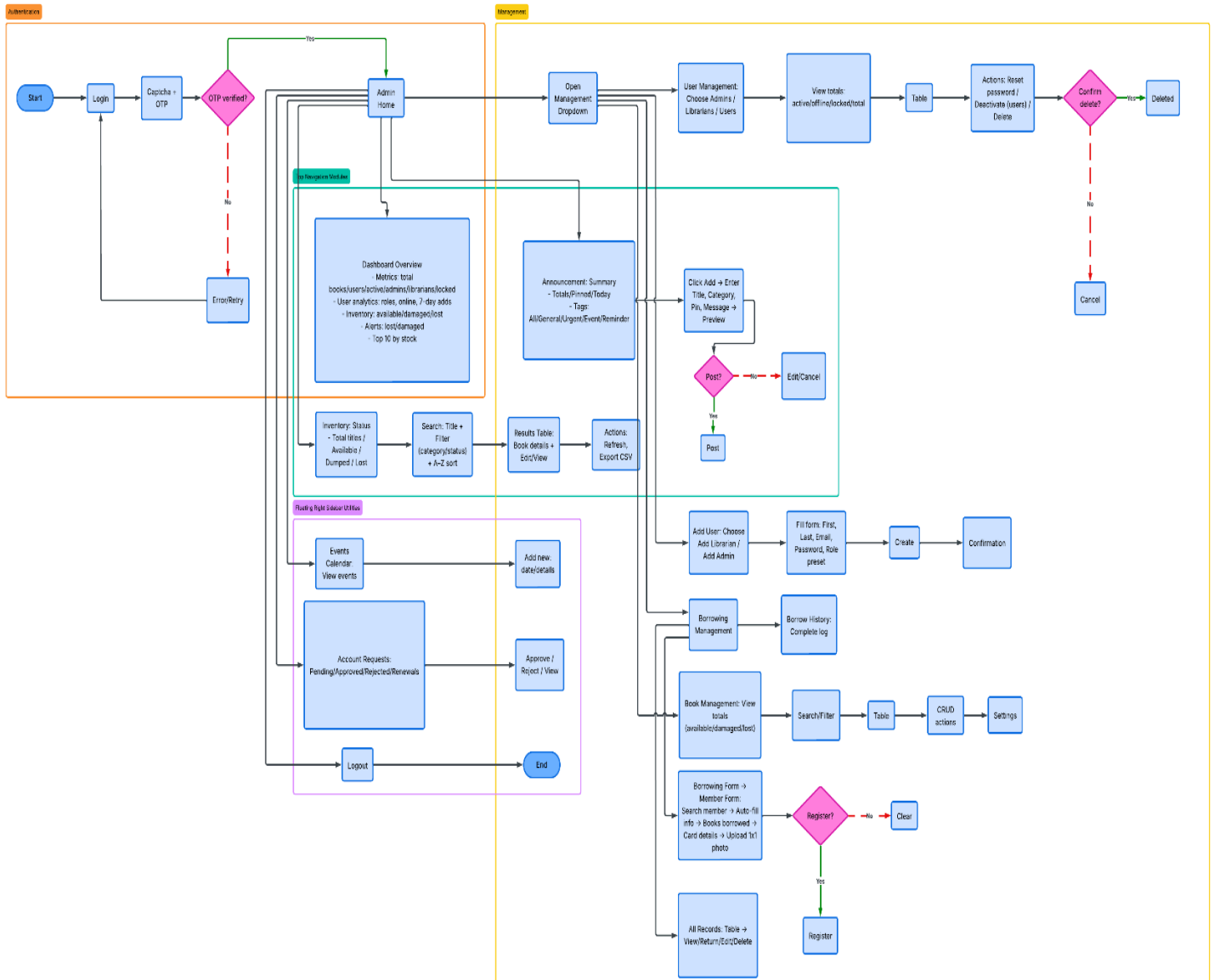
Member Flow:





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Admin Flow:



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Security Implementation

Control	Implementation	Purpose
HTTPS	SSL/TLS certificate	Encrypt traffic
RBAC	Flask session roles	Access restriction
Encryption	Fernet AES-128-CBC	Data at rest
OTP / MFA	Email-based 6-digit OTP	Multi-factor authentication
Brute Force Protection	Account lockout after 5 failed attempts (15 min)	Prevent credential stuffing
CAPTCHA	Math & trivia challenge on login	Bot prevention
Password Policy	Bcrypt hashing + complexity regex	Secure credential storage
Password Reset	Signed time-limited token (1 hr)	Secure recovery flow
Input Sanitization	HTML tag stripping + length limits	XSS / overflow prevention
Approval Workflow	Admin approval before account activation	Unauthorized access prevention
File Upload Validation	Extension whitelist + size limits	Malicious file prevention
Session Tracking	last_seen timestamp + auto-offline	Session awareness
Audit / Error Handling	DB rollback on exception + error handlers	Data integrity & accountability
Backup	Scheduled executable backup	Data recovery

3.4 Tools and technologies Used

- Frontend: HTML, CSS, JavaScript/Bootstrap
- Backend: Python /Flask
- Database: Xampp/MySQL
- Development Tools: Visual Studio Code

3.5 Data Gathering Techniques

- Interview with librarians to understand book management and borrowing processes works.
- Checked existing records like book lists and borrowing logs to understand how information is stored and managed.
- Observed how books are manually managed in the library, including how they are recorded and borrowed.

3.6 Testing Procedure

- **System Testing** – Tested the whole system to make sure it works properly.
- **User Testing** – Let users try the system to see if it is easy to use and understand.

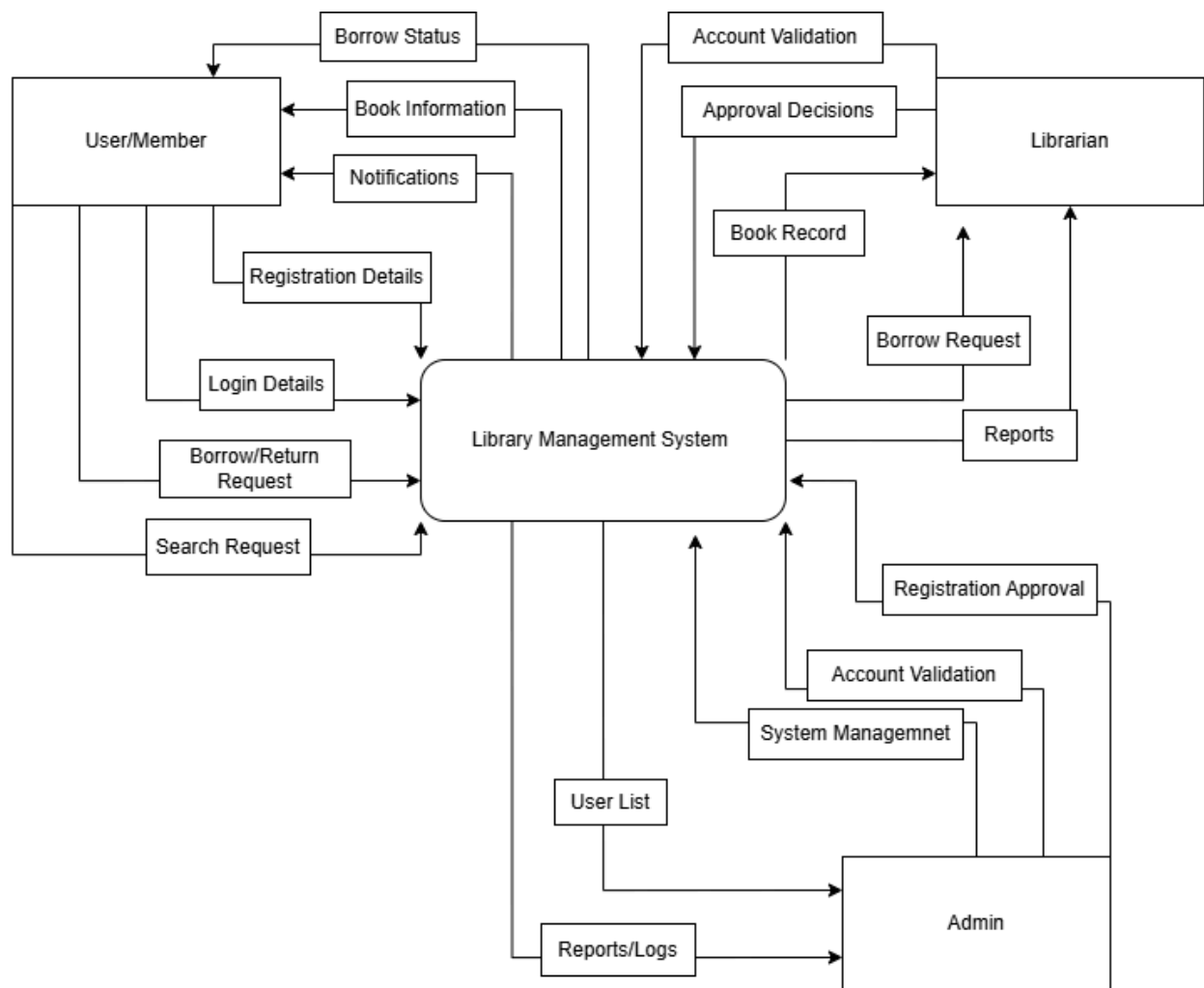
3.7 Evaluation Criteria

The system will be evaluated based on the following criteria:

- Accuracy of books status
- Efficiency
- Usability
- Reliability
- Functionality
- Availability

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Data Flow Diagram (DFD – Level 0)



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Entities:

Users (id, username, firstname, lastname, otp_code, password, created_at, failed_attempts, is_locked, lock_until, otp_expiry, status, last_seen, role, address, phone_number, reset_token, reset_token_expiry, age, sex, is_active, is_approved)

user_appearance (id, user_id, theme, language, updated_at)

user_favorites (id, user_id, book_id, created_at)

user_search_history (id, user_id, query, searched_at)

books (id, title, author, isbn, category, genre, publisher, subtitle, published_date, description, page_count, language, thumbnail_url, api_source, is_borrowable, created_at, updated_at)

book_inventory (id, book_id, total_copies, available_copies, damaged_copies, lost_copies, shelf_location, status, created_at)

book_returns (id, card_id, return_date, processed_by, created_at)

book_return_items (id, return_id, card_book_id, book_title, book_author, book_isbn, qty_returned)

events (id, title, description, event_date, start_time, end_time, location, image, created_at, category)

library_cards (id, card_type_category, user_id, firstname, lastname, phone_number, address, date_issued, date_return, renewal1_checked, renewal1_date, renewal2_checked, renewal2_date, card_type, valid_until, photo_path, registered_by, created_at, updated_at)

library_card_books (id, card_id, book_id, book_title, book_author, book_isbn, quantity, added_at)

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notification_reads (id, user_id, announcement_id, read_at, dismissed)

account_requests (id, user_id, username, request_type, reason, status, reviewed_by, admin_note, created_at, card_id, renewal1_checked, renewal1_date, renewal2_checked, renewal2_date)

announcements (id, title, body, category, author, pinned, created_at, updated_at)

Relationships:

- A User can have a Library Card
- A User can make an Account Request.
- An Admin checks Account Requests.
- An Admin can register Library Cards.
- Librarians can handle Book Returns.
- A User can read Announcements.
- A User can favorite Books.
- A User can have Search History.
- A User has Appearance Settings.
- A Library Card can borrow Books.
- A Book has an Inventory.
- A Book has one Inventory record.

Use Case Diagram

Actors:

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- Users
- Borrowers
- Admin
- Librarians

Use Cases:

Users / Member

- Explore Books
- Sign Up
- Login
- Browse Books
- Search Books
- Borrow Book
- Add to Favorites
- View Borrowed Books
- Renew Book
- View Notifications
- View Events
- Manage Profile
- Edit Profile
- Change Password

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- Adjust Settings
- Request Account Deletion
- Logout

Librarian

- Login
- View Dashboard
- View Member Records
- Manage Member Records
- Borrowing Management
- Process Renewals
- View Borrow History
- Book Management (Add/Edit/Delete)
- Manage Inventory
- View Alerts
- Manage Announcements
- Search/Filter Books
- Export Data (CSV)
- Logout

Admin

- Login

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- View Dashboard
- View Analytics
- Manage Users (Admin/Librarian/Member)
- Add User (Admin/Librarian)
- Reset Passwords
- Deactivate/Delete Accounts
- Borrowing Management
- Process Borrow Forms
- View Borrow History
- Book Management (CRUD)
- Manage Inventory
- Manage Announcements
- View Reports
- Handle Account Requests
- Approve/Reject Requests
- Manage Events Calendar
- Export Data

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Reference

- <https://www.rosemet.com/agile-methodology-steps/>
- Akor, S. O., Nongo, C., Udofot, C., & Oladokun, B. D. (2024). Cybersecurity Awareness: Leveraging Emerging Technologies in the Security and Management of Libraries in Higher Education Institutions . *Southern African Journal of Security*, 2, 14 pages .
<https://doi.org/10.25159/3005-4222/16671>
- Aregbesola, A., & Nwaolise, E. L. (2023). *Securing digital collections: Cyber security best practices for academic libraries in developing countries. Library Philosophy and Practice (e-journal)*. <https://digitalcommons.unl.edu/libphilprac/7822/>
- **Breeding, M. (2024).** *Libraries under cyberattack. Computers in Libraries*, 44(2).
<https://www.infotoday.com/cilmag/mar24/Breeding--Libraries-Under-Cyberattack.shtml>
- Espiritu, P. G. G., & Jocson, J. C. (2023). Navigating Cybersecurity Challenges in The Era of Digital Transformation: Threats and Mitigation Strategies in The Philippines. *International Journal of Progressive Research in Science and Engineering*, 4(11), 10–23.
Retrieved from <https://journal.ijprse.com/index.php/ijprse/article/view/994>
- Ayereby, Manouan Pierre-Marius, "Overcoming Data Breaches and Human Factors in Minimizing Threats to Cyber-Security Ecosystems" (2018). *Walden Dissertations and Doctoral Studies*.
<https://scholarworks.waldenu.edu/cgi/viewcontent.cgi?article=7442&context=dissertation>

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- "Exploring Cybersecurity Awareness and Training Strategies To Protect Information Systems and Data" (2020). *Walden Dissertations and Doctoral Studies*. 8902.
<https://scholarworks.waldenu.edu/cgi/viewcontent.cgi?article=10174&context=dissertations>
- National Privacy Commission. (2016). *Implementing rules and regulations of the Data Privacy Act of 2012 (Republic Act No. 10173)*.
<https://privacy.gov.ph/implementing-rules-regulations-data-privacy-act-2012/>
- Rick Anderson, Gwen Evans, and Jason Ensor, "Cybersecurity in the Library: Steps You Can Take Now to Make Your Users and Your Institutions Safer." *Proceedings of the IATUL Conferences*. Paper
1. <https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=2368&context=iatul>
- Swaminathan, S. (2024). *Digital transformation in library management systems*. BPAS Journal of Library and Information Science.
<https://bpasjournals.com/library-science/index.php/journal/article/view/2237>
- Singh, R., & Kumar, A. (2025). *Role of public libraries in the digital era: A comparative study*. ResearchGate.
https://www.researchgate.net/publication/394114500_Role_of_Public_Libraries_in_the_Digital_Era_A_Comparative_Study
- Lewis, J., & Lipkowitz, J. (2025). *Cybersecurity challenges in modern libraries*. Journal of Library Administration.
<https://www.tandfonline.com/doi/abs/10.1080/01616846.2025.2585761>

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

- Raghavaiah, K., & Karnati, C. (2025). *The importance of cybersecurity in academic libraries in the modern age*. ResearchGate.
https://www.researchgate.net/publication/387858933_The_Importance_of_Cybersecurity_in_Academic_Libraries_in_Modern_Age
- Library Systems Development: Utilization, Usability and Impact for Private Schools in Negros Occidental, Philippines Stephanie L. Lampaza¹ ; Mary Joy P. Parrenas² ; Angele D. Suarez³ ; Kristine T. Soberano, Ph. D.
<https://doi.org/10.47760/ijcsmc.2025.v14i01.002>
- Aregbesola, A., & Nwaolise, E. L. (2023). *Securing digital collections: Cybersecurity best practices for academic libraries in developing countries*. *Library Philosophy and Practice (e-journal)*.
<https://digitalcommons.unl.edu/libphilprac/7822/>

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